

Debugging Exercise

Task-1

Corrected Code:

```
<form id="userForm" onsubmit="validate()">
  <input type="text" id="username">
  <button type="submit">Submit</button>
</form>

<script>
function validate() {
  let name = document.getElementById("username").value;
  if (name == "") {
    alert("Name required");
    return false;
  }
  return true;
}
</script>
```

Explanation: The form was not submitting because the validation function did not control the submit event. Adding onsubmit="validate()" and returning true allows the form to submit.

Task-2

Corrected Code:

```
<p class="msg">Hello User</p>
<style>
.msg {
  color: blue;
  font-size: 20px;
}
</style>
```

Explanation: ' : ' was missing after color.

Task-3

Corrected Code:

```
<button id="btn">Click Me</button>
<script>
  document.getElementById("btn").onclick = function () {
    alert("Clicked");
  };
</script>
```

Explanation: The statement for selecting an element with ID attribute is document.getElementById("");

Task-4

Corrected Code:

```
let age = 16;
if (age >= 18) {
  console.log("Eligible");
} else if (age < 18) {
  console.log("Not Eligible");
}
```

Explanation: age === 18 is neglected

Task-5

Corrected Code:

```
function check() {
  let email = document.getElementById("email").value;
  console.log(email);
  if (email === "") {
    alert("Email required");
  }
}
```

Explanation: '.value' is required to get the values of any input field

Task-6

Corrected Code:

```
.card {
  width: 300px;
  padding: 20px;
}
```

Explanation: 'px', 'rem', 'em', '%' or any units should be mentioned along with specified number

Task-7

Corrected Code:

```
let pwd = "admin123";
let confirmPwd = "admin123";
if (pwd == confirmPwd) {
  console.log("Match");
} else {
  console.log("Mismatch");
}
```

Explanation: ' ' space given in confirmPwd variable and == check for only value of datatype so space gives a mismatch

Task-8

Corrected Code:

```
.alert {  
  visibility: visible;  
}
```

Explanation: The alert message was hidden because visibility: hidden was applied, preventing it from being visible even when triggered.

Task-9

Corrected Code:

```
console.log("Start");  
alert("Hello")  
console.log("End");
```

Explanation: ‘ “ ’ was missing in console.log(“End”) statement.

Task-10

Corrected Code:

```
function submitForm() {  
  if (terms.checked == false) {  
    alert("Accept terms");  
    return false;  
  }  
  return true;  
}
```

Explanation: return false and return true helps in form submission only when it is valid.

Task-11

Corrected Code:

```
<input type="text" id="mobile" />  
<button onclick="validate()">Submit</button>  
<script>  
  function validate() {  
    let mobile = document.getElementById("mobile").value;  
    if ((mobile.length === 10)) {  
      alert("Valid");  
    } else {  
      alert("Invalid");  
    }  
  }  
</script>
```

Explanation: Assignment operator ‘ = ’ has been used instead of ‘ === ’ comparison operator

Task-12

Corrected Code:

```
function validateName() {  
  let name = document.getElementById("policyName").value;  
  if (name == "") {  
    alert("Policy holder name required");  
  }  
}
```

Explanation: '.value' is required to retrieve the values given by the user in input fields

Task-13

Corrected Code:

```
<select id="plan">  
  <option>Select Plan</option>  
  <option>Health</option>  
  <option>Life</option>  
</select>  
<button onclick="checkPlan()">Apply</button>  
<script>  
  function checkPlan() {  
    let plan = document.getElementById("plan").value;  
    if ((plan == "Select Plan")) {  
      alert("Please choose a plan");  
    }  
  }  
</script>
```

Explanation: Assignment operator '=' has been used instead of comparison operator '=='

Task-14

Corrected Code;

```
let policyNumber = 123456;  
document.getElementById("policy").innerHTML = policyNumber;
```

Explanation: Incorrect variable('policyNo') used while assigning.

Task-15

Corrected Code:

```
let claimAmount = "abc";  
  
if (Number.isNaN(Number(claimAmount))) {  
  alert("Invalid claim");  
} else {  
  alert("Valid claim");  
}
```

```
}
```

Explanation: The validation was incorrect because isNaN() performs implicit type conversion. Using Number.isNaN(Number(value)) ensures accurate numeric validation.

Task-16

Corrected Code:

```
let policyType = "Health";  
if (policyType == "Health") {  
  console.log("Health Policy");  
}
```

Explanation: policyType was compared to 'health' where it should be compared with 'Health'

Task-17

Corrected Code:

```
let policies = ["Life", "Health", "Vehicle"];  
let list = document.getElementById("list");  
  
policies.forEach(function (policy) {  
  list.innerHTML += `<li>${policy}</li>`;  
});
```

Explanation; Considering the list is a ordered or unordered list elements of the array should be added as an element

Task-18

Corrected Code

```
let premium = "5000";  
if (Number.isNaN(Number(premium))) {  
  console.log("Invalid premium");  
} else {  
  console.log("Valid premium");  
}
```

Explanation: The validation was incorrect because isNaN() performs implicit type conversion. Converting the value to a number and using Number.isNaN() ensures correct validation.